Year 13 Maths - Applications of vectors

Similar questions - Variable force and velocity

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 In this question **i** and **j** are unit vectors due east and due north. A particle P of mass 2 kg starts from rest at the origin at $t = 0$ and moves on a smooth horizontal surface under a variable horizontal force $\mathbf{F} = ((6-2t)\mathbf{i} + 4t\,\mathbf{j})$ N, for $t \geqslant 0$.

(a) Find the value of t at which $\mathbf{F}$ has no **i** component. **[1]**

(b) Find the speed of P at the instant when it is moving due north. **[5]**

(c) Find the distance of P from the origin at that instant. **[5]**

2 In this question **i** and **j** are unit vectors due east and due north. A particle P of mass 0.5 kg moves so that at time t seconds, $t \geqslant 0$, its velocity is $\mathbf{v} = \left((1-2t)\mathbf{i} + (2t^2 + t - 13)\mathbf{j}\right)$ $\mathrm{m\,s^{-1}}$.

(a) Show that P is never stationary. **[2]**

(b) Find the acceleration of P at time t. **[1]**

(c) Find the magnitude of the force acting on P when it is moving in the direction of the vector $-2\mathbf{i} + \mathbf{j}$. **[5]**

(d) When $t = 1$, P is at the point with position vector $\frac{1}{6}\mathbf{j}$. Find the bearing of P from the origin when $t = 1.5$. **[5]**

Question 1

$\mathbf{a} = \frac{1}{2}\mathbf{F} = (3-t)\mathbf{i} + 2t\,\mathbf{j}$

(a) $6 - 2t = 0$ gives $t = 3$

(b) $\mathbf{v} = \int \mathbf{a}\,dt = \left(3t - \frac{1}{2}t^2\right)\mathbf{i} + t^2\mathbf{j}$ (no constant, as it starts from rest). Moving due north when $3t - \frac{1}{2}t^2 = 0$, so $t = 6$. Then $\mathbf{v} = 36\mathbf{j}$, speed $36\ \mathrm{m\,s^{-1}}$

(c) $\mathbf{r} = \left(\frac{3}{2}t^2 - \frac{1}{6}t^3\right)\mathbf{i} + \frac{1}{3}t^3\mathbf{j}$. At $t = 6$: $\mathbf{r} = 18\mathbf{i} + 72\mathbf{j}$. Distance $= \sqrt{18^2 + 72^2} = \sqrt{5508} = 74.2$ m

Question 2

(a) The **i** component is zero only when $t = \frac{1}{2}$, where the **j** component is $0.5 + 0.5 - 13 = -12 \neq 0$. So $\mathbf{v} \neq \mathbf{0}$ for all t

(b) $\mathbf{a} = -2\mathbf{i} + (4t+1)\mathbf{j}$

(c) Moving in the direction of $-2\mathbf{i} + \mathbf{j}$ needs $\dfrac{1-2t}{2t^2+t-13} = -2$, so $4t^2 = 25$ and $t = 2.5$ (where $\mathbf{v} = -4\mathbf{i} + 2\mathbf{j}$). $\mathbf{F} = 0.5\mathbf{a} = -\mathbf{i} + 5.5\mathbf{j}$, magnitude $\sqrt{1 + 30.25} = 5.59$ N

(d) $\mathbf{r} = (t - t^2)\mathbf{i} + \left(\frac{2}{3}t^3 + \frac{1}{2}t^2 - 13t + c\right)\mathbf{j}$. Using $\mathbf{r}(1) = \frac{1}{6}\mathbf{j}$ gives $c = 12$. At $t = 1.5$: $\mathbf{r} = -0.75\mathbf{i} - 4.125\mathbf{j}$, which is south-west of O. The bearing is $180^\circ + \arctan\dfrac{0.75}{4.125} = 190^\circ$ (3 s.f.)